OTF Trading Module – Functional Specification (MVP)

1. Purpose

The purpose of the OTF Trading module is to provide a simple position management and mark-to-market (MtM) environment for OTC/OTF electricity trading.

The first version (MVP) is intentionally limited to:

- Weekly BASE products only
- Manual trade entry
- Open position tracking
- Daily MtM valuation
- Basic P&L monitoring

The system is not intended to generate trading signals, forecasts, risk metrics, VaR calculations, or automated trading decisions in this phase.

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2. Scope

Supported products:

- BASE Week products only

Examples:

- BASE_W-27-26
- BASE_W-28-26
- BASE_W-29-26

The system must support both:

- LONG positions
- SHORT positions

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3. Navigation Structure

The OTF Trading section shall contain two pages:

3.1 Trades

Trade registration and maintenance.

3.2 MtM

Mark-to-market valuation of all open positions.

No additional pages are required in MVP.

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4. Trades Page

Purpose

Store all executed transactions and maintain a complete trading blotter.

Trade Entry Form

Required fields:

Field	Description
Transaction Date	Date when trade was executed
Product	Weekly product code
Position	LONG or SHORT
MW	Contract size in MW
Trade Price	Execution price (PLN/MWh)

The following fields should be calculated automatically:

Field	Description
Delivery Start	Derived from product code
Delivery End	Derived from product code
Hours	Number of delivery hours
Volume (MWh)	MW × Hours
Notional Value	Volume × Trade Price

Trade Table

Display all trades.

Columns:

Column
Trade ID
Transaction Date
Product
Position
MW
Delivery Start
Delivery End
Hours
Volume (MWh)
Trade Price
Notional Value
Status (Open / Closed)

Actions

Supported actions:

- Create Trade
- Edit Trade
- Delete Trade

Trade closing functionality is not required in MVP.

Positions are considered open until delivery completion.

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5. MtM Page

Purpose

Provide daily valuation of all open positions.

This page is the primary operational view for the trader.

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6. Synthetic Price Methodology

The system shall calculate a synthetic market price for each open weekly product.

The methodology must remain intentionally simple in MVP.

Before Delivery Starts

Synthetic Price = Trade Price

Therefore:

MtM = 0

No unrealized P&L exists before delivery begins.

This avoids introducing market data dependencies in MVP.

During Delivery Period

The weekly product consists of multiple delivery hours.

Some hours may already have realized market prices, while others remain undelivered.

Synthetic Price shall be calculated as:

Synthetic Price = (Realized Value + Remaining Value) / Total Hours

Where:

Realized Value = Sum of realized Fixing1 prices for completed hours

Remaining Value = Trade Price × Remaining Hours

In MVP:

Future hours shall be valued using the original trade price.

No forecasting models are used.

After Delivery Ends

Synthetic Price becomes:

Average Fixing1 price over all delivery hours.

The position becomes fully realized.

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7. Mark-to-Market Calculation

For LONG positions:

MtM = (Synthetic Price − Trade Price) × Volume

For SHORT positions:

MtM = (Trade Price − Synthetic Price) × Volume

Where:

Volume = MW × Hours

Result:

MtM is expressed in PLN.

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8. MtM Table

Display all open positions.

Columns:

Column
Product
Position
MW
Delivery Start
Delivery End
Trade Price
Synthetic Price
Hours
Realized Hours
Remaining Hours
MtM

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9. Summary Section

At the top of the MtM page display:

Metric
Total MtM
Number of Open Positions

No additional portfolio metrics are required in MVP.

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10. Assumptions

The MVP intentionally excludes:

- Forward market prices
- Fair value models
- Forecasting models
- Edge Score
- Trading signals
- VaR
- Greeks
- Portfolio optimization
- Automated trade recommendations

These capabilities may be added in future releases.

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11. Future Development Roadmap

Potential future extensions:

Phase 2:

- Monthly products
- Quarterly products
- Calendar products

Phase 3:

- Forward market valuation
- Forecast-based synthetic pricing

Phase 4:

- Trading signals
- Fair value engine

- Portfolio analytics
- Risk management

The MVP should be designed in a way that allows future expansion without major redesign.